Retrieval-Augmented Generation (RAG) Architecture

Page 1: Fundamental Concepts and Vector Stores

1. Vector Embeddings represent words and passages as dense mathematical vectors.

2. ChromaDB is an open-source, lightweight embedding database designed for AI apps.

3. HNSW (Hierarchical Navigable Small World) allows sub-linear approximate nearest neighbor search

4. Cosine similarity evaluates the angle between vectors rather than magnitude.

Local LLM Execution with Ollama and Llama 2

Page 2: Inference and Prompt Grounding

1. Ollama serves models via a unified REST API on port 11434.

2. Llama 2 utilizes 7 billion parameters with Grouped-Query Attention (GQA).

3. Prompt engineering provides strict system instructions preventing hallucinations.

4. Grounded generation synthesizes answers exclusively from retrieved context blocks.